

KDevelop Setup Guide

How to open, build, and run this project's three implementations (cpp, opencl, reference) from inside KDevelop.

1 Open the project

File → **Open Project**, then select the root `meson.build`:

Path	<project-root>/meson.build
------	----------------------------

2 Configure & build

1. Right-click the project name in the **Projects** tool view (left sidebar — open it via **View** → **Tool Views** → **Projects** if it's not visible) → **Open Configuration**. Confirm the Meson build directory is `<project-root>/build`, backend `ninja`. Apply/OK.
2. **Click directly on the project node** in the Projects sidebar to select it — the Build action stays greyed out until a project is actively selected there.
3. **Build** → **Build Project** (hammer icon, or `Ctrl+F8`). Wait for it to finish in the build output pane.

3 Set up the three launch configs

Run → **Configure Launches...**, then add each of these three (types noted below). Every field not listed can be left at its default.

projection_center_cpp · type: Compiled Binary

Mode	select the "Executable" radio button (not "Project Target")
Executable	<project-root>/build/forward_search_pipeline
Arguments	--data data/proj_shepplogan128.hdf5 --search-kernel forward_search/kernels/forward_search.cl --resample-kernel forward_search/kernels/resample.cl --output-pose /tmp/kdev_cpp_pose.h5 --output-data /tmp/kdev_cpp_resampled.hdf5
Working dir	<project-root>

projection_center_openc1 · type: Script Application

Interpreter	<python3>
Script	<projection-center>
Arguments	pipeline --backend openc1 --data data/proj_shepplogan128.hdf5 --output-pose /tmp/kdev_openc1_pose.json --output-data /tmp/kdev_openc1_resampled.hdf5
Working dir	<project-root>

projection_center_reference · type: Script Application

Interpreter	<python3>
Script	<project-root>/reference/run_reference_pipeline.py
Arguments	--data data/proj_shepplogan512.hdf5 --output-pose /tmp/kdev_reference_pose.json -- output-data /tmp/kdev_reference_resampled.hdf5
Working dir	<project-root>

4 Run one

1. **Run** → **Current Launch Configuration**, pick which of the three to make active.
2. **Run** → **Execute** (or Shift+F9). A terminal window opens showing that program's output.

✓ **Timing:** cpp and openc1 finish in seconds. Reference takes about **85 minutes** — that's normal, just wait.

